

CODING CHALLENGE - JAVA

HOSPITAL MANAGEMENT SYSTEM

Patient table:

	patientId	firstName	lastName	dateOfBirth	gender	contactNumber	address
▶	101	Ananya	Rao	1996-05-14	Female	9876543210	12 MG Road, chennai, TamilNadu
	102	Rahul	Verma	1989-11-23	Male	9123456780	45 Anna Nagar, Chennai, Tamil Nadu
	103	Divya	Srinivasan	1982-03-09	Female	9000012345	78 Residency Road, Coimbatore, Tamil Nadu
	104	Karan	Malhotra	1993-08-02	Male	9998877665	15 Banjara Hills, Trichy, Tamil Nadu
	105	Sneha	Kumar	1998-12-30	Female	9789456123	23 Salt Lake City, pondicherry, Tamil Nadu
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Doctor table:

	doctorId	firstName	lastName	specialization	contactNumber
▶	201	Aarav	Sharma	Cardiologist	9812345670
	202	Meera	Iyer	Dermatologist	9823456710
	203	Rohan	Raj	Orthopedic	9834567820
	204	Priya	Das	Pediatrician	9845678930
	205	Vikram	singh	Neurologist	9856789040
*	NULL	NULL	NULL	NULL	NULL

Appointment table:

	appointmentId	patientId	doctorId	appointmentDate	description
▶	301	101	201	2025-04-10	Routine heart check-up
	302	102	202	2025-04-11	Skin rash consultation
	303	103	203	2025-04-12	Knee pain evaluation
	304	104	204	2025-04-13	Child vaccination
	305	105	205	2025-04-14	Migraine and neurological exam
*	NULL	NULL	NULL	NULL	NULL

```
package main;
```

```
import java.util.List;
import dao.HospitalServiceImpl;
import entity.Appointment;
```

```
public class MainModule {
    public static void main(String[] args) {
```

```
        HospitalServiceImpl service = new HospitalServiceImpl();
```

```

//fetch an apointment by id
System.out.println("---- Fetch Appointment By ID ----");
Appointment apptById=service.getAppointementById(301);
if (apptById != null) {
System.out.println(apptById);
} else {
System.out.println("No appointment found with the given ID.");
}

```

☑ Database connection established successfully.

---- Fetch Appointment By ID ----

Appointment Details:

Appointment ID : 301

Patient ID : 101

Doctor ID : 201

Date : 2025-04-10

Description : Routine heart check-up

```

//to fetch the appointments for a patient
System.out.println("---- Fetch Appointments For Patient ----");
List<Appointment>patientAppointments =service.getAppointmentsForPatient(101);
for (Appointment a:patientAppointments) {
System.out.println(a);
}

```

☑ Database connection established successfully.

---- Fetch Appointments For Patient ----

Appointment Details:

Appointment ID : 301

Patient ID : 101

Doctor ID : 201

Date : 2025-04-10

Description : Routine heart check-up

```

//to fetch apointemnets for a doctor
System.out.println("---- Fetch Appointments For Doctor ----");
List<Appointment>doctorAppointments=service.getAppointmentsForDoctor(203);
for (Appointment a:doctorAppointments) {
System.out.println(a);
}

```

☑ Database connection established successfully.

---- Fetch Appointments For Doctor ----

Appointment Details:

Appointment ID : 303

Patient ID : 103

Doctor ID : 203

Date : 2025-04-12

Description : Knee pain evaluation

//to schedule a new appointment

System.out.println("---- Schedule New Appointment ----");

Appointment a1=new Appointment(306,101,201, "2025-04-20","Follow-up consultation");

boolean scheduled = service.scheduleAppointment(a1);

System.out.println("Appointment Scheduled:" + scheduled);

☑ Database connection established successfully.

---- Schedule New Appointment ----

Appointment Scheduled:true

	appointmentId	patientId	doctorId	appointmentDate	description
▶	301	101	201	2025-04-10	Routine heart check-up
	302	102	202	2025-04-11	Skin rash consultation
	303	103	203	2025-04-12	Knee pain evaluation
	304	104	204	2025-04-13	Child vaccination
	305	105	205	2025-04-14	Migraine and neurological exam
	306	101	201	2025-04-20	Follow-up consultation
*	NULL	NULL	NULL	NULL	NULL

//to update an appointment

System.out.println("---- Update Appointment ----");

a1.setDescription("Follow-up with new symptoms");

boolean updated = service.updateAppointment(a1);

System.out.println("Appointment Updated:" + updated);

☑ Database connection established successfully.

---- Update Appointment ----

Appointment Updated:true

	appointmentId	patientId	doctorId	appointmentDate	description
▶	301	101	201	2025-04-10	Routine heart check-up
	302	102	202	2025-04-11	Skin rash consultation
	303	103	203	2025-04-12	Knee pain evaluation
	304	104	204	2025-04-13	Child vaccination
	305	105	205	2025-04-14	Migraine and neurological exam
	306	101	201	2025-04-20	Follow-up with new symptoms
*	NULL	NULL	NULL	NULL	NULL

```
List<Appointment>patientAppointments =service.getAppointmentsForPatient(101);
for (Appointment a:patientAppointments) {
System.out.println(a);
}
```

<terminated> MainModule [Java Application] C:\Users\WELCOME\.p2\pool\plugins\org.ec

☑ Database connection established successfully.

Appointment Details:

Appointment ID : 301

Patient ID : 101

Doctor ID : 201

Date : 2025-04-10

Description : Routine heart check-up

Appointment Details:

Appointment ID : 306

Patient ID : 101

Doctor ID : 201

Date : 2025-04-20

Description : Follow-up with new symptoms

```
//cancel an appointment
```

```
System.out.println("---- Cancel Appointment ----");
```

```
boolean cancelled = service.cancelAppointment(306);
```

```
System.out.println("Appointment Cancelled:" + cancelled);
```

☑ Database connection established successfully.

---- Cancel Appointment ----

Appointment Cancelled:true

	appointmentId	patientId	doctorId	appointmentDate	description
▶	301	101	201	2025-04-10	Routine heart check-up
	302	102	202	2025-04-11	Skin rash consultation
	303	103	203	2025-04-12	Knee pain evaluation
	304	104	204	2025-04-13	Child vaccination
	305	105	205	2025-04-14	Migraine and neurological exam
*	NULL	NULL	NULL	NULL	NULL

}
}